Static NTK Obstructions in Classification and the Role of Feature Learning

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1 Introduction

Convolutional neural tangent kernels (CNTKs) provide a precise infinite-width kernel analogue of convolutional neural network training. In the lazy-training regime, the network function evolves as kernel gradient flow with a fixed kernel determined at initialization, making the training dynamics analytically tractable and reducing learning to kernel regression [9, 1, 11]. This viewpoint has led to a detailed theory of optimization and generalization for overparameterized neural networks. At the same time, empirical comparisons reveal a persistent gap between static NTK/CNTK predictors and finite-width neural networks trained with feature learning, especially on image classification benchmarks such as CIFAR-10 [1, 10]. This gap has motivated several explanations, including finite-width corrections, architectural refinements, data augmentation, preprocessing, label-aware kernels, and non-lazy feature learning [7, 8, 12, 5, 13].

In this paper, we focus on a complementary question. Rather than asking how to correct the CNTK so that it better matches a trained CNN, we ask what structural obstruction a fixed kernel faces in classification, and how did feature learning overcome the obstruction.

We first introduce standard kernel model as our benchmark(some of the analysis are original), then we come up with a formulation of the disaster caused by the phenomenon of local label mixing, finally we use numerical experiment to analyze how did feature learning escape from the obstruction by learning the kernel metric.

2 Benchmark: Static Kernel Model

In this section, we study the learning speed and generalization performance of kernel gradient flow.

We consider the nonparametric regression model

$$Y = f^*(X) + \xi,$$

where $X \sim \mu$ is the covariate on $\mathcal{X}$, and ξ is a conditionally mean-zero σ-sub-Gaussian noise. Given n i.i.d. samples $(x_i, y_i)_{i=1}^n$, our goal is to infer f^*. We assume the source condition

$$\|f^*\|_{[\mathcal{H}]^\nu} \le R, \qquad \nu > 0,$$

where $\mathcal{H}$ is the RKHS associated with a bounded kernel. Let $(\lambda_i)_{i \ge 1}$ be the eigenvalues of the corresponding kernel integral operator, and assume the polynomial eigenvalue decay

$$\lambda_i \asymp i^{-\beta}, \qquad \beta > 1.$$

2.1 Learning Speed

Let $K \in \mathbb{R}^{n \times n}$ be the empirical kernel matrix on training inputs $x_1, \dots, x_n$:

$$K_{ij} = k(x_i, x_j).$$

Write its eigendecomposition as

$$K = \sum_{j=1}^n \lambda_j v_j v_j^\top, \qquad \lambda_1 \ge \cdots \ge \lambda_n \ge 0,$$

with orthonormal eigenvectors v_j. For $m = 0, \dots, n$, define the projection onto the leading eigenspaces

$$P_m = \sum_{j \le m} v_j v_j^\top,$$

with $P_0 = 0$.

Consider the gradient flow estimator, using $\frac{1}{2n}\sum_{i=1}^n (y_i - \hat{f}(x_i))^2$ as the training loss, we can characterize the speed of the loss decaying.

Proposition 2.1. *The equation of static kernel gradient flow is*

$$\frac{d}{dt} f_t = -\frac{1}{n} K(f_t - y), \qquad f_0 = 0.$$

Let $r_t = y - f_t$. Then

$$r_t = \exp(-tK/n) y$$

and

$$\|r_t\|_2^2 = \sum_{j=1}^n \exp(-2\lambda_j t/n) \langle y, v_j \rangle^2.$$

Proof. This is a standard result, for proof see appendix A. □

Definition 2.2. For any nonzero vector $u \in \mathbb{R}^n$, define

$$E_u(m) = \frac{\|P_m u\|_2^2}{\|u\|_2^2}, \qquad T_u(m) = 1 - E_u(m) = \frac{\|(I - P_m) u\|_2^2}{\|u\|_2^2}.$$

For labels $y \in \{\pm 1\}^n$, we call $E_y(m)$ the cumulative spectral label energy and $T_y(m)$ the spectral label tail energy.

We can see that the loss decays when $E_y(m)$ is relatively big. Some literature already showed that when source condition ($\nu = 2$) is satisfied, $T_y(m)$ is relatively small([4, 3]), here we establish results for all $\nu > 0$.

Theorem 2.3. *Let $s \in \mathbb{R}^n$ be the evaluation vector $s_i = f(x_i)$, there exist a constant C independent of n such that*

$$\mathbb{P}(T_s(m) \leq C\lambda_m n^{\max(-1,-\nu)}) \to 1$$

where the randomness is from draw of $x_i (1 \leq i \leq n)$

Moreover, we can show that the energy concentrates in the first eigenvectors base on the results in [6].

Theorem 2.4. *Let $s \in \mathbb{R}^n$ be the evaluation vector $s_i = f(x_i)$, there exist a constant $c > 0$ independent of n such that*

$$\mathbb{P}(E_s(m) \geq c\lambda_m) \to 1$$

where the randomness is from draw of $x_i (1 \leq i \leq n)$

Since we can make the decomposition $y = s + \xi$, for ξ it is flat such that $[E_m(\xi)] \asymp \frac{m}{n}$ for subgaussian noises.

Theorem 2.5. *There exist constants $c, C > 0$, such that for every fixed m and every $t > 0$,*

$$\mathbb{P}\left(\left| E_\xi(m) - \sigma^2 \frac{m}{n} \right| \geq C\sigma^2 (\frac{\sqrt{mt} + t}{n}) \;\middle|\; x_1, \dots, x_n \right) \leq 2e^{-ct}.$$

For the proof of theorem 2.3, theorem 2.4, and theorem 2.5, see appendix B.

2.2 Generalization

Now we state the following minimax rate. For $\nu \geq 1$, see [2]; for $0 < \nu < 1$, see the proof of [14].

Proposition 2.6 (Minimax squared $L^2(\mu)$ risk)**.** *Under the assumptions above, when $nR^2/\sigma^2 \gtrsim 1$, the minimax squared $L^2(\mu)$ risk over the source ball is*

$$\inf_{\widehat{f}} \sup_{\|f^*\|_{[\mathcal{H}]^\nu} \leq R} \mathbb{E}\left[\|\widehat{f} - f^*\|^2_{L^2(\mu)} \right] \asymp R^{\frac{2}{\nu\beta+1}} \sigma^{\frac{2\nu\beta}{\nu\beta+1}} n^{-\frac{\nu\beta}{\nu\beta+1}}.$$

We comment that the generalization error is positively related to the RKHS norm(or the source norm) of f^*.

3 Label Mixing Obstruction for Classification Problem

In this section, we formulate the main obstruction caused by local label mixing and its consequence on training speed and generalization ability.

3.1 Obstruction

Define the kernel metric $d_k(x,y) = k(x,x) + k(y,y) - 2k(x,y)$.

Theorem 3.1. *Let k be a positive definite kernel with RKHS $\mathcal{H}_k$, and let $K \in \mathbb{R}^{n\times n}$ be the empirical kernel matrix. Write*

$$K = \sum_{j=1}^{n} \lambda_j v_j v_j^\top, \qquad \lambda_1 \geq \cdots \geq \lambda_n \geq 0.$$

Assume $\lambda_m > 0$, and let $P_m = \sum_{j\leq m} v_j v_j^\top$. Let $y \in \{\pm 1\}^n$, and define the unnormalized tail

$$\widetilde{T}_m(y) := \|(I - P_m)y\|_2^2.$$

Let $G = ([n], E, w)$ be a weighted undirected graph whose edges connect only opposite labels:

$$(i,j) \in E \quad \Longrightarrow \quad y_i \neq y_j.$$

Let $w_{ij} > 0$ and define

$$W_G := \sum_{(i,j)\in E} w_{ij}, \qquad u^\top L_G u = \sum_{(i,j)\in E} w_{ij}(u_i - u_j)^2, \qquad \Delta_G := \lambda_{\max}(L_G).$$

Assume $W_G > 0$, and define the average squared kernel-edge length

$$\bar{\rho}_G^2 := \frac{1}{W_G} \sum_{(i,j)\in E} w_{ij} d_k(x_i, x_j)^2.$$

Then

$$\widetilde{T}_m(y) \geq \frac{W_G}{\Delta_G} \left(2 - \bar{\rho}_G \sqrt{\frac{n - \widetilde{T}_m(y)}{\lambda_m}} \right)_+^2.$$

In particular,

$$\widetilde{T}_m(y) \geq \frac{W_G}{\Delta_G} \left(2 - \bar{\rho}_G \sqrt{\frac{n}{\lambda_m}} \right)_+^2.$$

Equivalently, the normalized spectral label tail energy satisfies the same lower bounds divided by $\|y\|_2^2 = n$.

Corollary 3.2 (Disjoint local collisions)**.** *Suppose there exist q_ρ disjoint opposite-label pairs (i_ℓ, j_ℓ), $\ell = 1, \ldots, q_\rho$, such that*

$$y_{i_\ell} \neq y_{j_\ell}, \qquad d_k(x_{i_\ell}, x_{j_\ell}) \leq \rho,$$

and no index appears in more than one pair. Then

$$\widetilde{T}_m(y) \geq \frac{q_\rho}{2} \left(2 - \rho \sqrt{\frac{n - \widetilde{T}_m(y)}{\lambda_m}} \right)_+^2.$$

In particular,

$$\widetilde{T}_m(y) \geq \frac{q_\rho}{2}\left(2 - \rho\sqrt{\frac{n}{\lambda_m}}\right)_+^2.$$

If $\lambda_m \geq n\rho^2$*, then* $\widetilde{T}_m(y) \geq q_\rho/2$*.*

3.2 Consequence

4 Role of feature Learning

5 Experiments

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